



Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : E. Ticketing Website

Group No : 02, CSE370 Lab Section : 18, Spring 2026		
ID	Name	Contribution
22241103	Adiyan Abdur Rahman	Login, Logout and Registration; Search and Filter Events; Role-based access, event display, user profile, booking history and other core features
23301353	Alvi Bin Mobarok	Admin Venue, Event and Discount Management; Enroll Event on Cart; Load User Balance
23301066	Mahir Azmol Shimanto	Apply Discount During Payment; Checkout; Feedback Section

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	3
3	ER/EER Diagram	4
4	Schema Diagram	5
5	Normalization	6
6	Frontend Development	7
7	Backend Development	10
8	Source Code Repository	13
9	Conclusion	13
10	References	13
11	ScreenShots	14

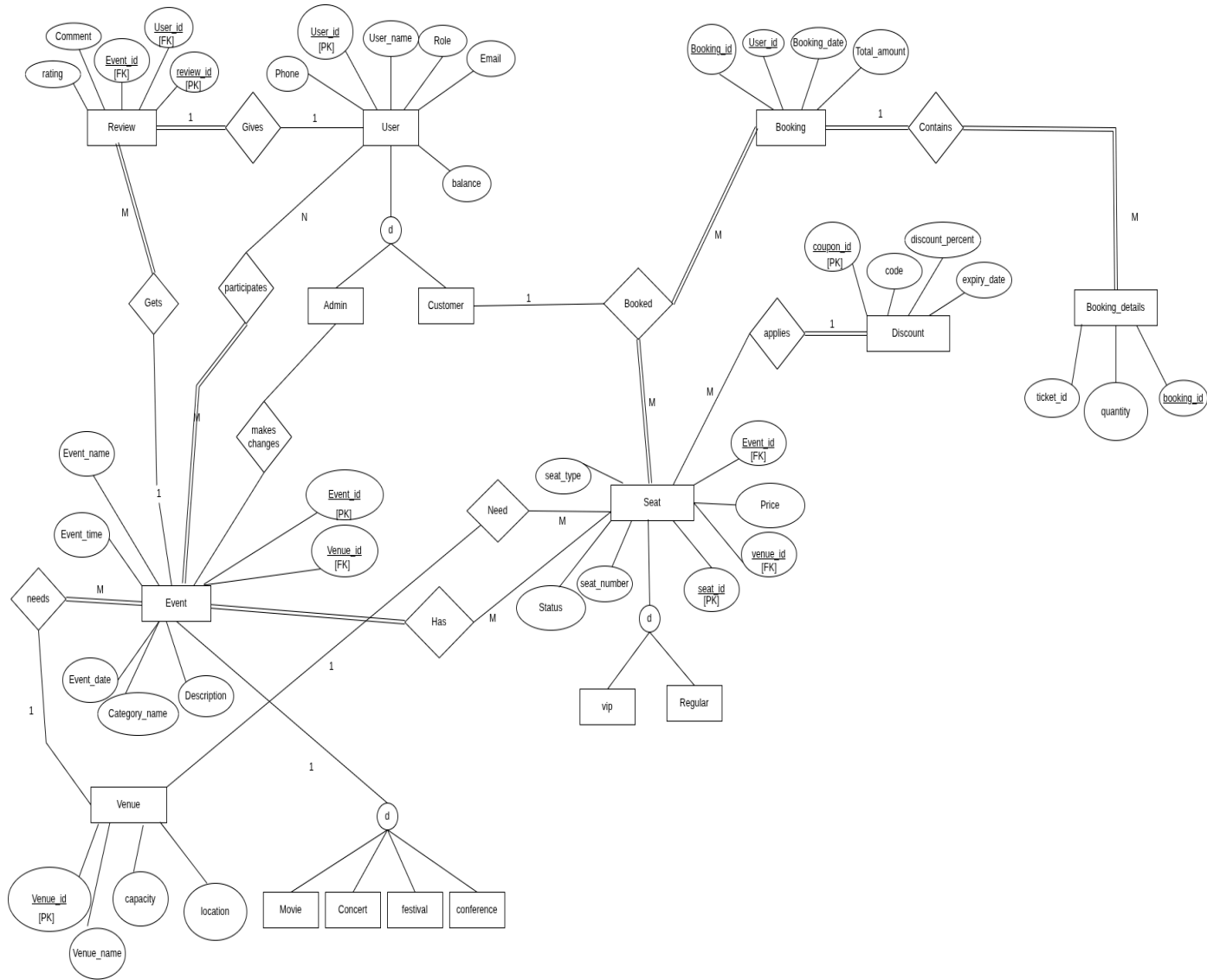
Introduction

E-Ticketing is a user-friendly, web-based ticket booking system designed to make event discovery and ticket purchasing simple, organized, and efficient. It allows users to register and log in to their accounts, browse different types of events such as movies, concerts, conferences, and festivals, and search or filter events based on category, date, or venue. The platform provides detailed event information, seat selection, ticket availability checking, and multiple ticket booking options to ensure a smooth booking experience. It also includes secure payment processing, discount and coupon code features, booking history, user profile management, and event review functionality. With role-based access for Admin and Customer, the system enables administrators to manage events, users, bookings, and platform activities effectively. By focusing on simplicity, security, and usability, the E-Ticketing website helps users book tickets conveniently while giving administrators proper control over the ticketing process.

Project Features

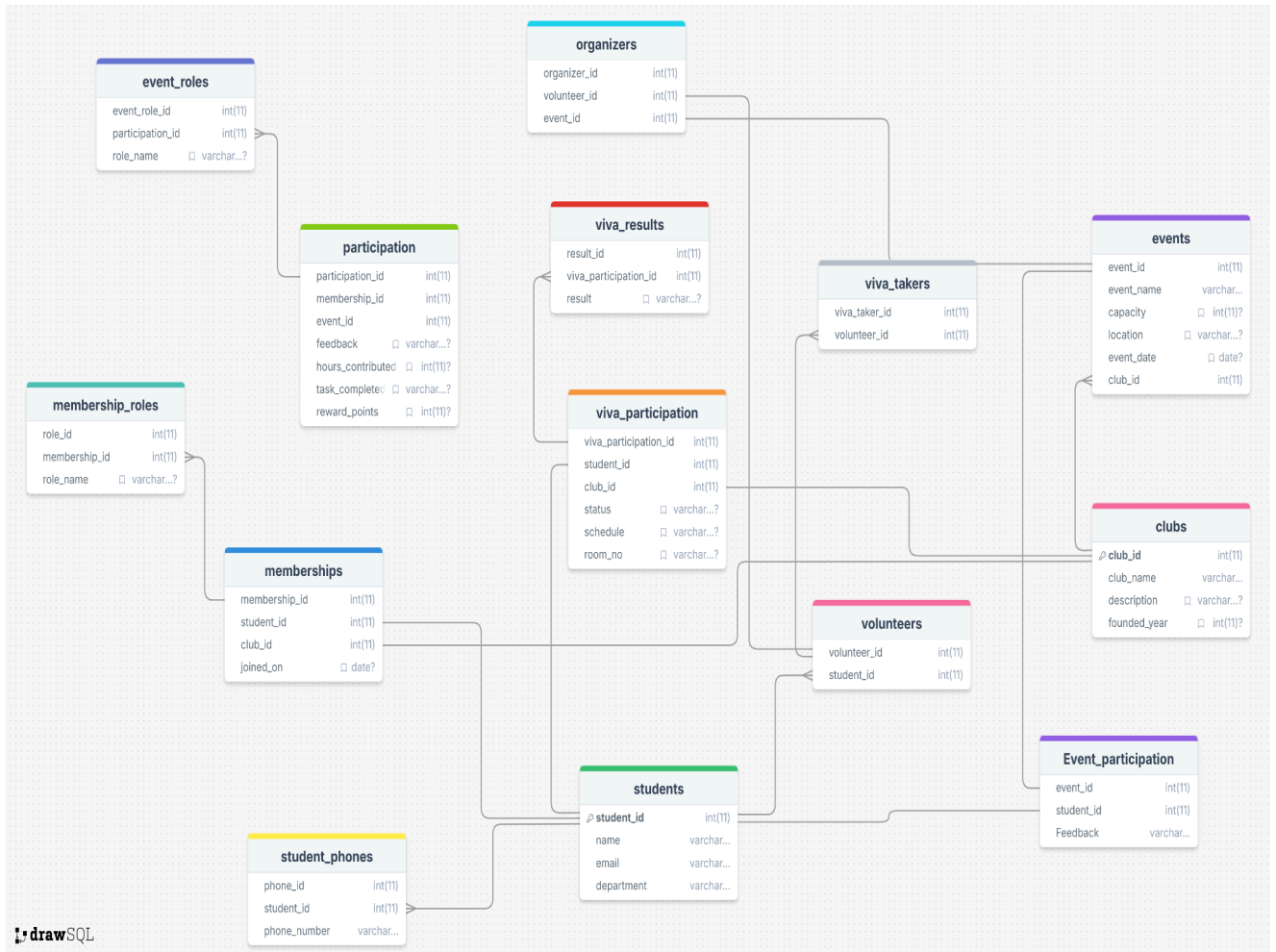
ID, Name	Features [3 per member]	
20101999, Adiyan Abdur Rahman	Ft 1	Log in and Log out /Registration
	Ft 2	Seat generation
	Ft 3	User profile event result fetching
23301353, ALVI BIN MOBAROK	Ft 1	As a admin adding Venue, Event, Discount
	Ft 2	Enroll event on Cart
	Ft 3	Load user ballance
23301066, Mahir Azmol Shimanto	Ft 1	Apply discount during payment
	Ft 2	Checkout
	Ft 3	Feedback section

ER/EER Diagram



For better view, kindly [CLICK HERE](#)

Schema Diagram



For better view, kindly [CLICK HERE](#)

Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Answer:

a. The converted database schema is in First Normal Form (1NF) because all tables have atomic columns, no repeating groups, and each table has a clearly defined primary key. For example, tables like users, events, venues, seats, bookings, discounts, and reviews store individual data values in separate columns. Also, the booking_details table handles multiple ticket information separately instead of storing repeated ticket data inside the bookings table.

b. The schema is also in Second Normal Form (2NF). Since all tables are already in 1NF and every non-key attribute fully depends on the whole primary key, there are no partial dependencies. Most tables have single-column primary keys, such as user_id, event_id, venue_id, seat_id, booking_id, and coupon_id, so partial dependency cannot occur. In tables like booking_details, attributes such as quantity depend on the full booking-ticket relationship, not only part of it.

c. The schema satisfies Third Normal Form (3NF) because there are no transitive dependencies among non-key attributes. All non-key columns depend directly on their table's primary key. For example, event details depend on event_id, venue details depend on venue_id, booking details depend on booking_id, and review details depend on review_id. Related data such as user, venue, event, seat, discount, and booking information are kept in separate tables using foreign keys, which reduces redundancy and avoids transitive dependency.

Frontend Development

The frontend of **TIX E-Ticketing Website** was developed using **HTML, CSS, JavaScript, and Laravel Blade**. The interface was designed to provide a clean, modern, and user-friendly experience for both **customers** and **admins**. The frontend connects with the PHP backend and MySQL database to display events, venues, ticket details, booking information, discounts, and user-specific data dynamically.

Contribution of ID : 22241103, Name : Adiyen Abdur Rahman

1. Landing Page and Public Event Display

- a. Home Page UI: Designed the main landing page with a modern hero section, navigation bar, login/register buttons, and featured event banner.
- b. Upcoming Events Section: Created event cards showing event images, category labels, event names, dates, locations, and “Buy Tickets” buttons.
- c. User Navigation: Implemented clear navigation options such as Home, Events, Login, Register, and Explore Events to make the website easy to use.

2. Event Browsing Interface

- a. Event Listing Page: Developed the customer-facing event browsing page where users can view available events in a card-based layout.
- b. Event Category Display: Added visual category badges such as Festival and Regular to help users identify event types quickly.
- c. Price Display: Designed the price section to show ticket prices clearly in BDT, making it easier for users to compare events.

3. Responsive and Visual Design

- a. Layout Design: Used clean spacing, card shadows, rounded corners, and consistent colors to create a professional user interface.
- b. Image Presentation: Added event images and promotional visuals to make the platform more attractive and engaging.
- c. Frontend Consistency: Maintained a consistent color theme, typography, button style, and spacing across the public pages.

Contribution of ID : 23301353, Name : ALVI BIN MOBAROK

1. Admin Dashboard Interface

- a. Admin Home UI: Designed the admin dashboard with separate cards for managing Venues, Events, and Discounts.
- b. Management Cards: Created large clickable cards with icons, titles, and short descriptions so admins can easily access important management features.
- c. Admin Navigation: Implemented a simple top navigation bar with the TIX logo and profile icon for admin users.

2. Event and Venue Management Pages

- a. Existing Events Display: Developed the admin event list interface showing event name, date, venue status, and seat generation option.
- b. Venue Review Page: Created the venue listing page where venues are displayed with image, name, location, and capacity.
- c. Admin-Friendly Layout: Designed the admin pages with a clean grid layout so that event and venue data can be viewed and managed easily.

3. Seat and Event Control UI

- a. Seat Generation Option: Added frontend action links such as “Click to generate seats” for admin event control.
- b. Event Information Cards: Designed event cards in the admin panel to present important event details in a readable format.
- c. Usability Focus: Ensured that admin tasks such as managing events, venues, and discounts are accessible from simple dashboard options.

Contribution of ID : 23301066, Name : Mahir Azmol Shimanto

1. Customer Interaction Interface

- a. User Profile Dropdown: Designed the profile dropdown that shows the user name, balance, load balance option, My Tickets, Settings, and Sign Out.
- b. Cart Indicator: Added a cart icon with ticket count display so users can easily track selected tickets.
- c. Customer Navigation: Implemented customer-side navigation between Events, Venues, cart, and profile options.

2. Event Details and Ticket Selection Modal

- a. Event Details Popup: Created a modal window to display event image, event type, event name, date, venue, description, ticket price, and available tickets.
- b. Ticket Quantity Control: Added plus and minus controls so users can increase or decrease ticket quantity before adding to cart.
- c. Add to Cart Button: Designed the “Add to Cart” button and ticket availability warning to guide users during booking.

3. Payment and Checkout UI

- a. Load Balance Modal: Designed the balance loading popup with card number, expiry date, CVC, and amount input fields.
- b. Checkout Page: Created the order summary page showing discount code input, subtotal, discount amount, total price, and confirm purchase button.
- c. Discount Code Interface: Implemented the coupon input and apply button to support discount-based ticket purchasing.

Backend Development

Contribution of ID : 22241103, Name : Adiyan Abdur Rahman

Contribution of ID: 20101999, Name: Adiyan Abdur Rahman

1. Login, Logout, and Registration

- a. User Registration: Implemented backend logic for creating new user accounts and storing user information in the database.
- b. Login System: Verified user email and password from the database and allowed valid users to access the system.
- c. Logout Functionality: Managed user sessions and safely logged users out from the system.

2. Search and Filter Events

- a. Filtered Queries: Wrote backend queries to search events by category, date, and venue.
- b. Dynamic Event Retrieval: Retrieved matching event records from the database and sent them to the frontend for display.
- c. Event Browsing Support: Ensured users can easily find relevant events based on their selected filter options.

3. Other Core Features

- a. Role-Based Access: Implemented backend logic to separate Admin and Customer access.
- b. Event Display: Retrieved event details, venue information, ticket price, and seat availability from the database.
- c. User Profile and Booking History: Managed user-related data such as profile information, previous bookings, and ticket records.

Contribution of ID : 23301353, Name : Alvi Bin Mobarok

1. Admin Management: Venue, Event, and Discount

- a. Venue Management: Implemented backend CRUD operations for adding and managing venue details such as venue name, location, and capacity.

b. Event Management: Created backend functionality for admins to add and manage event information including event name, date, time, category, description, and venue.

c. Discount Management: Developed backend logic for creating coupon codes, discount percentages, and expiry dates.

2. Enroll Event on Cart

a. Cart Handling: Implemented backend logic to add selected events and tickets to the user's cart.

b. Ticket Quantity Management: Managed ticket quantity and ticket type selected by the user before checkout.

c. Seat Availability Check: Checked ticket or seat availability before allowing the user to add an event to the cart.

3. Load User Balance

a. Balance Update: Developed backend functionality to load balance into a user account.

b. Payment Data Handling: Processed card input fields and amount data from the frontend form.

c. Database Update: Updated the user's balance in the database after successful balance loading.

Contribution of ID : 23301066, Name : Mahir Azmol Shimanto

Contribution of ID: 23301066, Name: Mahir Azmol Shimanto

1. Apply Discount During Payment

a. Coupon Validation: Implemented backend logic to check whether the entered discount code is valid or expired.

b. Discount Calculation: Calculated the discount amount based on the coupon percentage.

c. Total Amount Update: Updated the final payable amount after applying the discount.

2. Checkout

- a. Order Summary Processing: Calculated subtotal, discount, and total price for the selected tickets.
- b. Booking Confirmation: Created booking records in the database after confirming purchase.
- c. Booking Details Storage: Stored selected ticket information, quantity, and payment details in the booking-related tables.

3. Feedback Section

- a. Review Submission: Implemented backend logic for users to submit ratings and comments for events.
- b. Review Storage: Stored feedback data in the review table with user and event references.
- c. Review Retrieval: Retrieved user feedback and displayed it with the related event information.

Source Code Repository

[GITHUB](#)

[Google drive](#)

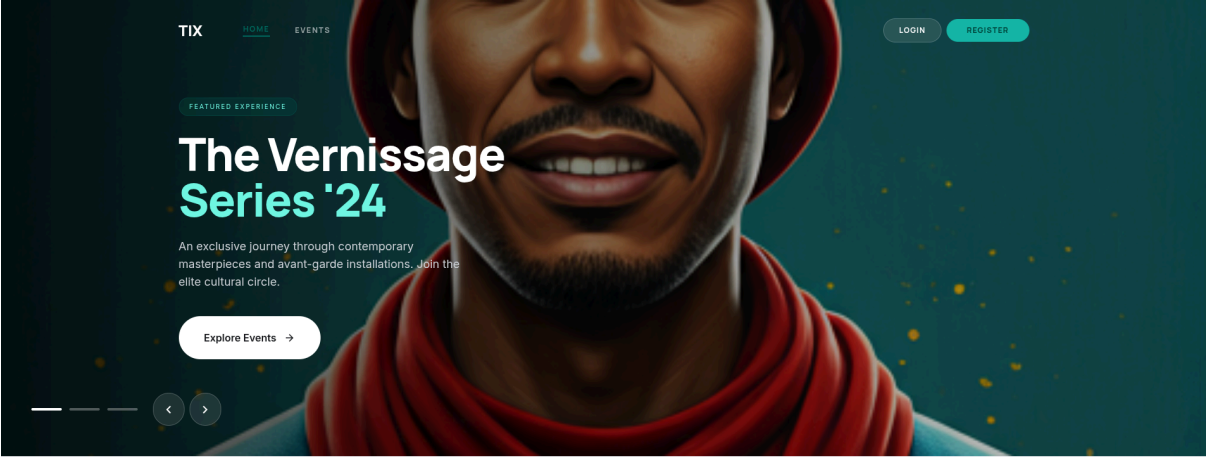
Conclusion

TIX E-Ticketing Website provides a simple and efficient platform for users to browse events, view event details, select tickets, apply discounts, and complete bookings through a structured checkout process. The system supports both customer and admin roles, allowing customers to manage their tickets and balance while admins can manage venues, events, discounts, and seat generation. By using PHP, MySQL, HTML, CSS, JavaScript, and Blade, the project combines frontend usability with backend database functionality. Overall, this project demonstrates how a database-driven web application can make event ticket booking more organized, secure, and user-friendly.

References

1. Mobarok, A. B. (2026). eticket-dbms-project. GitHub. Retrieved May 9, 2026, from <https://github.com/alvibinmobarok/eticket-dbms-project.git>
2. Rahman, A. A. (2026). GitHub Profile / Project Contribution. GitHub. Retrieved May 9, 2026, from <https://github.com/AdiyanAbdurRahman>
3. Mozilla. (n.d.). MDN Web Docs: HTML. Retrieved May 9, 2026, from <https://developer.mozilla.org/en-US/docs/Web/HTML>
4. Mozilla. (n.d.). MDN Web Docs: CSS. Retrieved May 9, 2026, from <https://developer.mozilla.org/en-US/docs/Web/CSS>
5. Mozilla. (n.d.). MDN Web Docs: JavaScript. Retrieved May 9, 2026, from <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
6. PHP Group. (n.d.). PHP Manual*. Retrieved May 9, 2026, from <https://www.php.net/manual/en/>
7. Oracle. (n.d.). MySQL Documentation. Retrieved May 9, 2026, from <https://dev.mysql.com/doc/>
8. Laravel. (n.d.). Laravel Documentation. Retrieved May 9, 2026, from <https://laravel.com/docs>

ScreenShots:



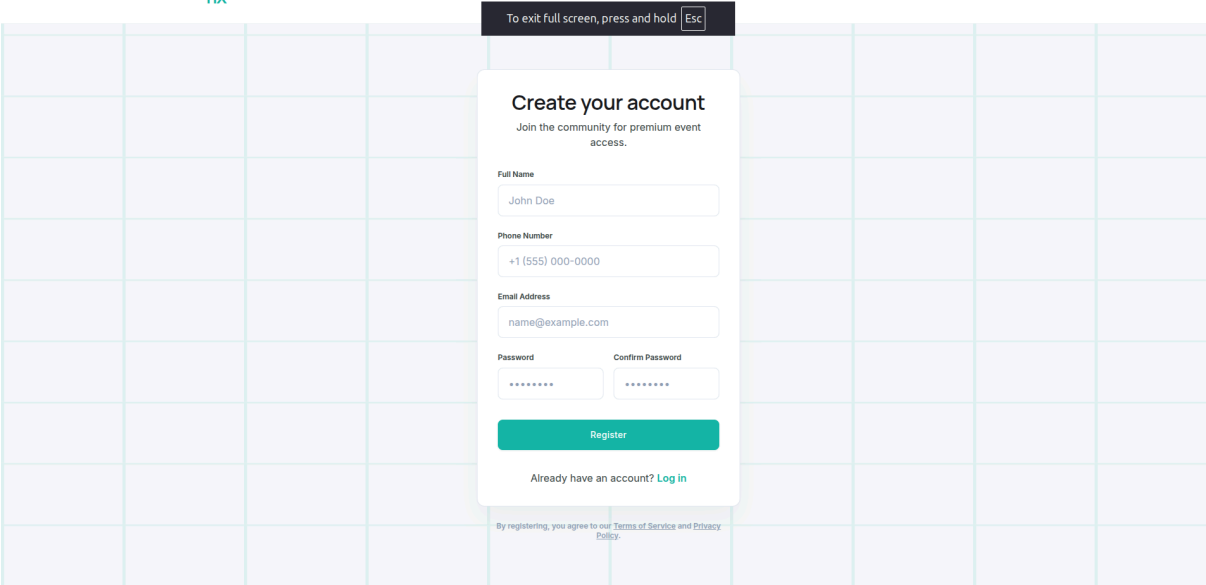
Upcoming Experiences

Curated selections from the world's most vibrant cultural hubs.

FILTER SORT



TIX





To exit full screen, press and hold **Esc**



Venues

Add and manage your venue destinations, capacities, and specifications.



Events

Curate experiences, manage ticketing, and upload event promotional assets.



Discounts

Create coupon codes, set discount percentages, and expiry dates.



To exit full screen, press and hold **Esc**

EVENT

Add Event



Event Name

Event Date



Event Time



Venue

Category

Description

Add

